

CGA Math

Spencer T. Parkin

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This paper documents the math needed to compose and decompose elements of CGA.

Foundation

For any blade B of CGA, it represents two geometries. These are all vectors x in the euclidean sub-algebra of CGA such that

$$p(x) \cdot B = 0,$$

or all such vectors with

$$p(x) \wedge B = 0,$$

where $p(x)$ is defined as the vector

$$p(x) = o + x + \frac{1}{2}x^2\infty.$$

We'll call the first of these the “dot representation” and the second, “wedge representation”. To convert from one to the other, you can multiply by I , the unit psuedo-scalar. This is because

$$p(x) \cdot B = 0 \iff p(x) \wedge BI = 0.$$

Of course, B and BI are duals of one another, so the conversion goes both ways.

Using blades to represent geometries, we get the following useful properties. If B_1 and B_2 are blades with $B_1 \wedge B_2 \neq 0$, then we have

$$\{x|p(x) \cdot B_1 \wedge B_2 = 0\} = \{x|p(x) \cdot B_1 = 0\} \cap \{x|p(x) \cdot B_2 = 0\},$$

as well as

$$\{x|p(x) \wedge B_1 \wedge B_2 = 0\} \supseteq \{x|p(x) \wedge B_1 = 0\} \cup \{x|p(x) \wedge B_2 = 0\}.$$

In this latter case, since there are only so many shapes in CGA, we can reasonably deduce that $B_1 \wedge B_2$ will be the geometry that fits those represented by B_1 and B_2 . You may, for example, find the sphere that fits a given circle and point, or the circle that fits 3 given points, and so on.

The shapes of CGA are either round or flat. If using the wedge presentation, a round can be converted into a flat by tacking on ∞ with the outer product. For example, if B represents a circle using the wedge representation, then $B \wedge \infty$ is the plane containing the circle, also in the wedge representation.

Compositions

For all geometry that follows, we let c and n denote euclidean vectors, and r denote a scalar. We always require that $|n| = 1$.

We are going to use the dot representation here instead of the wedge representation, and ignore the magnitude of B as long as it is non-zero.

Planes

Letting $B = n + (n \cdot c)\infty$, we have

$$\begin{aligned} 0 &= p(x) \cdot B \\ &= \left(o + x + \frac{1}{2}x^2\infty\right) \cdot (n + (c \cdot n)\infty) \\ &= -c \cdot n + x \cdot n \\ &= (x - c) \cdot n, \end{aligned}$$

which is clearly the equation for a plane centered at c with normal n .

Lines

Lines are a bit trickier. Here, let $i = e_1e_2e_3$ denote the unit psuedo-scalar of the 3D euclidean sub-algebra of CGA. Doing so, let $B = ni + (ni \cdot c)\infty$. We

then see that

$$\begin{aligned}
0 &= p(x) \cdot B \\
&= \left(o + x + \frac{1}{2}x^2\infty\right) \cdot (n - (c \cdot ni)\infty) \\
&= \left(o + x + \frac{1}{2}x^2\infty\right) \cdot (n + \infty \wedge (c \cdot ni)) \\
&= (o \cdot \infty)(c \cdot ni) + x \cdot ni - (x \cdot (c \cdot ni))\infty \\
&= -c \cdot ni + x \cdot ni - (x \cdot (c \wedge n)i)\infty \\
&= (x - c) \cdot ni - (x \wedge c \wedge n)i\infty.
\end{aligned}$$

Here, we must have $0 = (x - c) \cdot ni$ as well as $0 = x \wedge c \wedge n$. The first of these is clearly a line centered at c with normal n , which is what we're after, but what of the second? Well, we need only consider those values of x for which the first is satisfied. If $x = c$, we have a trivial case, and all is well. Otherwise, we have $x - c = \lambda n$ for some non-zero scalar λ , which means that $x \wedge c \wedge n = x \wedge c \wedge \frac{1}{\lambda}(x - c)$, which is clearly zero.

Spheres

Consider $B = (o + c + \frac{1}{2}(c^2 - r^2)\infty)$. We then have

$$\begin{aligned}
0 &= p(x) \cdot B \\
&= \left(o + x + \frac{1}{2}x^2\infty\right) \cdot \left(o + c + \frac{1}{2}(c^2 - r^2)\infty\right) \\
&= -\frac{1}{2}(c^2 - r^2) + x \cdot c - \frac{1}{2}x^2.
\end{aligned}$$

Rearranging this, we find that

$$\begin{aligned}
r^2 &= c^2 - 2(x \cdot c) + x^2 \\
&= (x - c)^2,
\end{aligned}$$

which is clearly the equation for a sphere.

Interestingly, we also find imaginary spheres in CGA with $B = (o + c + \frac{1}{2}(c^2 + r^2)\infty)$. Imaginary rounds are necessary in cases where an intersection is computed that can't actually exist.

Points

Points are simply the special case of spheres with $r = 0$. A point at c is simply $p(c)$. Note that these, technically, are round points, not to be confused with tangent points or flat points.

For a flat point, let

$$B = (n + (n \cdot c)\infty) \wedge (ni + (ni \cdot c)\infty),$$

which is clearly the intersection of a line and a plane. If you were to expand this wedge product out, things would cancel, and you'd be left with

$$B = i + ci \wedge \infty,$$

showing that indeed n is not applicable here.

Circles

Here, let

$$B = (n + (n \cdot c)\infty) \wedge \left(o + c + \frac{1}{2}(c^2 - r^2)\infty \right),$$

which is the intersection of a sphere centered on a plane. Use an imaginary sphere here to get an imaginary circle. Such circles form the intersection of, for example, two real spheres that are intersected, but that don't actually intersect.

Point-Pairs

Point-pairs may seem odd, but they're simply the 1-dimensional analog of spheres and circles. Here, let

$$B = (ni + (ni \cdot c)\infty) \wedge \left(o + c + \frac{1}{2}(c^2 - r^2)\infty \right).$$

This is simply the intersection of a line going through the center of a sphere. Similar to cricles, use an imaginary sphere here to get an imaginary point-pair. Such a point-pair would occur, for example, when you compute the intersection of a line and a sphere that don't actually intersect.

Tangent Points

These are the special case of circles or point-pairs when $r = 0$. Note that while it takes 4 points to fit a sphere, it only takes 2 tangent points, or 2 points and 1 tangent point, to do the same. You can get a tangent point by intersecting two spheres that touch in a single point.

Decomposition

Here we utilize the composition formulas derived earlier in order to extract the variables c , n and r from a given blade B . Since we can't ignore magnitude here, we need an extra scalar w to represent the “weight” of the geometry.

Transformations

Rotors

Motors

Transversions